

Relational Identity Structure (RIS): reproducibility correction and preliminary benchmark

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Abstract

Relational Identity Structure (RIS) is a prototype which derives node signatures from typed graph relations and can merge highly similar nodes. Earlier versions of this project reported benchmark, ablation, and scalability results without the datasets, scripts, configurations, or raw outputs needed to verify them. This paper withdraws those claims and replaces them with an executable, narrower evaluation. On fixed CompERBench entity-matching test sets, a 64-dimensional pre-merge RIS signature obtains F1 0.430 (Abt-Buy), 0.539 (Amazon-Google), and 0.991 (DBLP-ACM); a generic token-Jaccard baseline obtains 0.368, 0.559, and 0.993. We also evaluate actual online merges on the Leipzig Affiliations clustering benchmark: RIS reaches random-order F1 0.235 $\pm$ 0.011, below a greedy Jaccard profile baseline at 0.250 $\pm$ 0.008, and varies from 0.225 to 0.254 across four arrival orders. We then revise the policy with unsupervised IDF relation weights and summed evidence on merge. Chosen solely on calibration clusters, this version reaches F1 0.576 $\pm$ 0.014, exceeding both greedy Jaccard (0.250 $\pm$ 0.008) and exact weighted-cosine profile assignment (0.443 $\pm$ 0.011) on the fixed test stream. This is a one-dataset, offline-vocabulary result, not a general performance claim. The contribution is a falsifiable baseline: runnable code, raw metrics, checksums, and explicit limitations.

Keywords: entity resolution, record linkage, graph representation, reproducibility, structural equivalence

1. Correction and scope

RIS represents entities as graph nodes, forms a signature from neighboring relations, and offers a thresholded, irreversible merge. The premise is related to structural equivalence in network analysis [Lorrain and White, 1971], but it is a design hypothesis rather than a correctness guarantee.

The prior repository documentation and PDF reported results on synthetic customers, Cora and Amazon-Google; comparisons with rule-based, Dedupe, DeepMatcher, Ditto and GNN systems; ablations; HNSW latency; and memory or large-scale results. At audit time the repository had only the prototype and an example: no experiment directory, data, configuration, raw result, or runner supported those statements. They are withdrawn, not reconstructed.

This revision evaluates both a pairwise signature readout on labeled candidate pairs and one specified online merge policy. It does not establish general correctness, order-stability, utility, or scalability of automatic merging.

2. Prototype

For a node v with neighboring relations (u, lambda, w) , the code uses $\text{phi}(v) = \text{normalize}(\sum w * \text{encode}(u, \text{lambda}))$. encode is a deterministic pseudorandom vector keyed by the relation label and neighboring node ID. Pair score is $\text{cosine}(\text{phi}(a), \text{phi}(b))$. In normal library operation, similarly typed nodes with at least two relations can merge when their score exceeds a threshold. Merges are irreversible and change the graph.

Automatic merging is disabled for the pairwise study. Candidate-pair test sets do not specify a ground-truth temporal merge stream or a final gold partition, so allowing unrelated online merges would make that evaluation ill-defined. Section 5 separately evaluates a fully specified online policy on data with cluster ground truth.

3. Protocol

3.1 Data

We use the public fixed validation/test entity-matching tasks distributed by CompERBench [Primpeli and Bizer, 2020].

Task	Domain	Records	Validation pairs / positives	Test pairs / positives
Abt-Buy	products	2,173	1,439 / 220	710 / 109
Amazon-Google	products	2,404	1,696 / 261	836 / 128
DBLP-ACM	bibliography	4,910	9,387 / 447	4,624 / 220

The runner downloads the official records.zip, gs_val.csv, and gs_test.csv; SHA-256 values for the actual inputs are in results/benchmark_results.json.

3.2 Graph and selection

The adapter creates a record node per source record. It applies Unicode NFKC and case folding to every non-ID value, extracts generic `\w+` tokens, creates one field-token node per distinct (field, token), and links record to token with `has_token::<field>::<token>`. There are no domain lexicons, fuzzy-string rules, external knowledge, or labels in graph construction.

We compute 64-dimensional signatures and predict a candidate match only when cosine score is strictly greater than tau. Token allocation is sorted before IDs are assigned: because the encoder uses neighboring IDs, unordered set iteration would otherwise make the output process-dependent.

For each method and dataset, tau is selected from 0.00, 0.01, ..., 1.00 by validation F1. Ties prefer higher precision, then the higher threshold. The value is frozen before scoring test pairs. Thus the procedure uses validation labels for calibration; it is not a label-free thresholding claim.

3.3 Baseline and metrics

The only rerun comparator is exact-token Jaccard over the same field-prefixed tokens and with the same validation procedure. It is a sanity baseline, not a state-of-the-art claim. We report fixed-test pairwise precision, recall and F1; we do not report systems that were not executed in this repository.

4. Results

Dataset	Method	tau	Precision	Recall	F1	TP	FP	FN
Abt-Buy	RIS signature	0.28	0.353	0.550	0.430	60	110	49
Abt-Buy	Token Jaccard	0.14	0.312	0.450	0.368	49	108	60
Amazon-Google	RIS signature	0.33	0.575	0.508	0.539	65	48	63
Amazon-Google	Token Jaccard	0.10	0.471	0.688	0.559	88	99	40
DBLP-ACM	RIS signature	0.55	0.986	0.995	0.991	219	3	1
DBLP-ACM	Token Jaccard	0.41	1.000	0.986	0.993	217	0	3

RIS is marginally below Jaccard on DBLP-ACM and Amazon-Google. It is higher on Abt-Buy, though its absolute F1 is only 0.430. The results are heterogeneous and do not establish an entity-matching advantage. With this adapter the RIS signature is more plausibly a lossy random projection of token overlap.

5. Streaming merge evaluation

The public Leipzig Affiliations benchmark contains 2,260 affiliation strings, 330 gold clusters, and a complete perfect mapping. Connected components of the mapping define the reference clusters. A deterministic hash assigns whole gold clusters to calibration (70 clusters; 497 records) or test (260 clusters; 1,763 records), preventing records of the same entity from crossing the split.

For each arriving record, RIS loads its affiliation-token relations, finds the highest-scoring active record node, and merges only that candidate when its score exceeds tau. The calibration stream selects tau from 0.05 to 0.95 in 0.05 increments; the test uses that fixed threshold. We compare the policy to a greedy Jaccard token-profile merge and a thresholded Jaccard connected-components algorithm. All use the same NFKC, case-folded generic tokens.

Method	tau	Lexical F1	Random F1 mean +/- sd	Range	Random P	Random R
RIS online merge	0.45	0.242	0.235 +/- 0.011	0.225–0.254	0.266	0.211
Greedy Jaccard profile	0.15	0.263	0.250 +/- 0.008	0.238–0.260	0.277	0.227
Jaccard connected components	0.55	0.068	0.068 +/- 0.000	0.068–0.068	0.036	0.615

RIS is below the greedy baseline and varies by 0.029 F1 across shuffled arrival orders. That variation is a policy property, not sampling noise: the records and test partition do not change. The connected-components result makes a separate point: transitive closure is order-invariant but creates many false links. This single evaluation does not show a dynamic advantage for RIS.

5.1 Weighted evidence revision

The first streaming evaluation identifies two limitations of the prototype: common relation tokens have the same influence as discriminative tokens, and a merge retains only one edge when multiple members share a relation. We introduce two changes that preserve the relational model. First, record-token edge weight is IDF, $\log((N+1)/(df+1))+1$, computed once from the complete unlabeled raw corpus. Second, a merge **sums** weights of identical relations, so a cluster signature preserves how many of its members support a relation.

Using only the calibration split, we select dimension from {128, 256, 512, 1024}, aggregation from {maximum, sum}, and tau from 0.30–0.70 in 0.05 steps. The selected configuration is 256 dimensions, sum aggregation, and tau=0.35 (calibration F1=0.794). It is frozen before the test stream. We introduce a stronger control: the same greedy profile policy scored by exact IDF-weighted cosine, using the identical unlabeled IDF values.

Method	Lexical F1	Random F1 mean +/- sd	Range	Random P	Random R
RIS weighted signature, summed relations	**0.567**	**0.576 +/- 0.014**	0.552–0.589	0.567	0.587
Greedy weighted-cosine profile	0.468	0.443 +/- 0.011	0.429–0.458	0.484	0.408
Greedy Jaccard profile	0.263	0.250 +/- 0.008	0.238–0.260	0.277	0.227
Jaccard connected components	0.068	0.068 +/- 0.000	0.068–0.068	0.036	0.615

On this fixed test, summed relational evidence improves mean F1 by 0.133 over the exact weighted-cosine greedy control. The gain is consistent with the mechanism: after a correct merge, support for characteristic relations grows and subsequent candidates are scored against that accumulated context. The 0.037 F1 random-order range shows this does not solve path dependence.

This is not a strict future-blind streaming experiment: IDF is estimated from the entire unlabeled corpus before arrivals begin. Nor is it evidence of cross-domain generalization. It is a reproducible, bounded result for the specified Affiliations policy only.

6. Limitations

The streaming studies measure one single-source dataset and use an offline unlabeled IDF vocabulary; they do not generalize to other domains, strict future-blind streams, temporal drift, adversarial changes, HNSW recall or latency, memory consumption, scalability, learned signatures, or external ER systems. The prototype cannot undo a false merge. It should be treated as a research artifact, not used for unattended production deduplication.

Future end-to-end work needs a specified record-arrival stream, update and merge policy, alias policy, and merge-level or partition-level ground truth. Claims about any such work should ship the runner, configuration, raw outputs, fixed splits, and input checksums.

7. Reproducibility and conclusion

From the repository root run:

```
python experiments/run_benchmarks.py --download --output-dir results
python experiments/run_streaming_benchmark.py --download --output-dir results/streaming
python experiments/run_weighted_streaming_benchmark.py --download --output-dir
results/streaming_weighted
```

The runner produces raw JSON metrics, a Markdown report, selected thresholds, input checksums, and graph sizes. It requires Python and NumPy; optional hnswlib is not used in this pairwise evaluation. With the same code and data, scores reproduce exactly; wall-clock graph-build time may vary.

RIS remains a research prototype, but the current evidence now includes a bounded dynamic merge advantage: on one fixed, offline-vocabulary clustering stream, weighted relation signatures with evidence-preserving merges exceed two greedy lexical controls. It still performs inconsistently in the earlier pairwise tasks, and the streaming policy remains order-sensitive. The proper claim is therefore specific and testable, not universal: accumulated, rarity-weighted relational context can improve this merge policy on the Affiliations benchmark. Cross-domain and genuinely online validation remain necessary.

References

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